Roshan Hegde

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Education

University of Illinois at Urbana-Champaign May 2027
B.S. in Computer Science, Grainger College of Engineering *GPA: 3.91/4.0*
B.S. in Mathematics, College of Liberal Arts & Sciences
Highland Park High School May 2023
GPA: 4.69/4.0

Relevant Coursework

Courses: Algorithms, Operating Systems, Computer Architecture, Programming Languages and Compilers, Cryptography, Data Structures, Probability and Statistics for Computer Science, Applied Linear Algebra

Skills

Languages: C, C++, Rust, Go, Java, Kotlin, Python, Typescript
Tools: Git, Docker, Linux, Nix, Vim

Work Experience

Software Engineering Intern, Chicago IL | *Olam Food and Ingredients | Azure, python* May – August 2025

- Analyzed current utilization of cloud assets and identified opportunities to improve cost efficiency by 12%
- Developed a data analysis service to automate VM performance reporting to identify over-allocated resources
- Proposed a plan to improve resource cost-effectiveness to the infrastructure team

University Of Illinois Urbana-Champaign | *Operating Systems Course Assistant | C, GDB* August 2025 - Present

- Developed and reviewed programming assignments for students taking the course
- Answered both conceptual and implementation questions while hosting office hours

MechMania, Champaign IL | *Lead Infrastructure Developer | Go, Rust, Python, Nix* August 2024 – Present

- Built a multithreaded web-socket server in Go interacting with AWS services including databases and cloud storage
- Implemented a custom shared memory IPC protocol for multi-language support
- Leveraged Nix and bubble-wrap to run arbitrary code without vulnerabilities

Projects

LumonOS: Operating System | *C, Unix, RISC-V, QEMU, GDB* January – August 2025

- Created fully functional, RISC-V, Unix-based pre-emptive operating system with round robin scheduling policy.
- Built a virtio block driver, cryptography driver, and entropy driver
- Implemented support for pipes between processes and argument passing according to POSIX standard
- Placed 2nd in the Operating System design competition hosted by the ECE department

nix-tinker | *Rust, Cargo, Nix* May 2024 – Present

- Developed an open-source tool for rapid prototyping of Nix configuration files, streamlining development workflow
- Crafted an intuitive CLI for managing file links with the Nix store, significantly enhancing user efficiency
- Leveraged Nix Flakes to guarantee reproducible and consistent build environments across diverse systems

Awards

1st place, Hackillinois '24, '25 | *Rust, Cargo* February 2024 and 2025

- Competed with over 1,100 other students at Illinois' largest Hackathon
- Developed a full stack developer tool intended for rapid prototyping of data pipelines
- Defined a strongly typed JSON based protocol for ease of extensibility and versatility
- Built a developer tool used to universally initialize a basic development environment for over 30 languages

1st place, MechMania '24, '25 | *Java, Python* September 2023 and 2024

- Implemented A* pathfinding with a modified Manhattan distance heuristic
- Fine-tuned heuristics to pilot dogfighting planes, leveraging predictive modeling to gain a competitive edge